An exact half-line completion for the polynomial counterexample

Consider the single C1 source

$$(d,\lambda,n,m,C,A,D,\alpha,\beta)=\left(359,0,0,\frac{75}{16},60,416,\frac{41}{2},\frac{831}{2},\frac{151}{2}\right).$$

The scaling by 359 is only to simplify the rational coefficients. Every matrix and selector below divided by 359 gives a certificate for the same source normalized to $d=1$. Define

$$P(v)=\begin{pmatrix}75v^2/4 & -75v/4\\ -75v/4 & 4155/4+1020v^2\end{pmatrix},$$
$$N(v)=\begin{pmatrix}1040+240v^2 & 800v\\ 800v & 800v^2\end{pmatrix},$$
$$E(v)=\begin{pmatrix}1664+4044v^2 & -2380v\\ -2380v & 1662-718v^2\end{pmatrix}.$$

For $v\ge 0$, an exact quadratic completion is

$$\boxed{J_+(v)=\frac{1}{64}\begin{pmatrix}3043v-2109v^2 & 3875v^2\\ 1403+23874v+2306v^2 & 815v+43455v^2\end{pmatrix}.} \tag{1}$$

Put

$$G(v)=\begin{pmatrix}P(v) & J_+(v)\\ J_+(v)^T & N(v)\end{pmatrix},\qquad H(v)=\begin{pmatrix}P(v)+E(v) & -E(v)-J_+(v)\\ -E(v)-J_+(v)^T & N(v)+E(v)\end{pmatrix},$$

and let G_j,H_j denote their coefficients of v^j. Here are rational Gram matrices proving $G(v),H(v)\succeq 0$ on the whole half-line. Set

$$B_G=\frac{1}{256}\begin{pmatrix}0 & 0 & 0 & 0\\ -4800 & -48015 & -27346 & 3260\\ 12172 & 186862 & -21340 & 204800\\ 0 & 0 & 0 & 0\end{pmatrix}, \tag{2}$$

$$B_H=\frac{1}{18496}\begin{pmatrix}-6910952 & 240550 & 5277344 & 1292578\\ -41117866 & -4908665 & 35528504 & 2316913\\ 3043153 & 1489795 & -7316324 & 276862\\ 36194938 & -415582 & -30578223 & -3325523\end{pmatrix}. \tag{3}$$

Then define the four constant matrices

$$S_G=\begin{pmatrix}G_0 & B_G\\ B_G^T & G_2\end{pmatrix},\qquad T_G=G_1-B_G-B_G^T,$$
$$S_H=\begin{pmatrix}H_0 & B_H\\ B_H^T & H_2\end{pmatrix},\qquad T_H=H_1-B_H-B_H^T. \tag{4}$$

All four are positive semidefinite. An exact rational LDL^T certificate is specified as follows, including the singular directions. Use unpivoted symmetric elimination in the displayed coordinate order, with unit diagonal in L. At a positive pivot d_i, set $L_{ji}=R_{ji}/d_i$ and update $R_{jk}\leftarrow R_{jk}-R_{ji}R_{ik}/d_i$ for $j,k>i$. At a zero pivot the remaining row is identically zero and is left unchanged. Starting from each matrix in (4), the following table gives all zero pivot positions and exact integer lower bounds on the other pivots, in order; each positive pivot is strictly less than its bound plus one. Positions are numbered starting at one.

matrix	zero pivot positions	integer lower bounds on positive pivots
S_G	$1,4,8$	1038, 1039, 16, 382, 162
S_H	8	1664, 2700, 1039, 639, 1562, 214, 67
T_G	$1,3,4$	375
T_H	4	747, 483, 430

These are rational inequalities, not floating point bounds. For example, the rank-one matrix is explicitly

$$T_G=\frac{5335}{128}\begin{pmatrix}0\\3\\-2\\0\end{pmatrix}\begin{pmatrix}0 & 3 & -2 & 0\end{pmatrix}.$$

The full rational entries of L and D are also supplied in `SIMPLE_SOURCE_HALFLINE_EXACT_WITNESS.json`. The zero-pivot rows and the equality $M = LDL^T$ hold exactly for each of the four matrices. Their ranks are respectively $5, 7, 1, 3$.

For $R_v = [I_4 \;\; vI_4]$, direct coefficient comparison now gives

$$G(v) = R_v S_G R_v^T + vT_G, \qquad H(v) = R_v S_H R_v^T + vT_H. \tag{5}$$

Every coefficient and every PSD assertion in this certificate can be checked with rational arithmetic alone. The accompanying standard-library script `check_simple_source_exact_witness.py` independently reconstructs P, N, E, compares all quadratic coefficients in (5), recomputes the four LDL^T factorizations, and checks the exact contact kernels. It does not read the numerical diagnostic used to find the witness.

Restoration of the complete parent. Let $U = u^2$ and $a, b \in \mathbb{R}^2$. Define

$$\Phi(U, v; a, b) = (1+U)a^T P(v)a + U(1+U)b^T N(v)b + U(a-b)^T E(v)(a-b).$$

The exact identity

$$\Phi = \begin{pmatrix} a \\ Ub \end{pmatrix}^T G(v) \begin{pmatrix} a \\ Ub \end{pmatrix} + U \begin{pmatrix} a \\ b \end{pmatrix}^T H(v) \begin{pmatrix} a \\ b \end{pmatrix}$$

proves positivity for every $U \geq 0$, every a, b, and $v \geq 0$. The canonical state substitution

$$W = a_1 + Ub_1, \quad X = a_2 + Ub_2, \qquad Y = u(a_1 - b_1), \quad Z = u(a_2 - b_2)$$

restores the full physical parent as $F = (1+u^2)\Phi$. For $u \neq 0$ this substitution is invertible. Polynomial continuity gives $u = 0$, and homogenization in both angle pairs gives the projective endpoints.

For negative v, set $T = \mathrm{diag}(1, -1)$ and define

$$J_-(v) = TJ_+(-v)T \qquad (v < 0). \tag{6}$$

The exact parity identities $M(-v) = TM(v)T$ for $M = P, N, E$ show that both completion matrices for J_- are congruent to those at $-v > 0$. At $v = 0$ use $J_+(0)$, whose certificate is literal in (5). Thus all testing angles and all state directions are covered for this fixed source.

Why the polynomial obstruction is avoided. The selector in (6) together with (1) is piecewise quadratic. For instance, its upper-left entry is $(3043|v| - 2109v^2)/64$ and its upper-right entry is $3875v|v|/64$. It is not a polynomial on the real line. A global polynomial selector, if one existed, could be averaged under $v \mapsto -v$ and reduced by the endpoint zero rows to

$$J(v) = \begin{pmatrix} pv^2 & 0 \\ rv & sv^2 \end{pmatrix}.$$

At the exact contact $U = 1/4, v = 2, a = (17, 2)^T, b = (12, -18)^T$, the zero quadratic form forces

$$J(2)b = -4P(2)a, \qquad J(2)^T a = -N(2)b/4,$$

which uniquely give $(p, r, s) = (-100, 2000, 1200)$. Their $G(1)$ principal submatrix on (a_2, b_1, b_2) is

$$\begin{pmatrix} 8235/4 & 2000 & 1200 \\ 2000 & 1280 & 800 \\ 1200 & 800 & 800 \end{pmatrix}, \qquad \det = -412640000 < 0.$$

Thus the exhibited half-line certificate succeeds for a source excluded by every global polynomial selector. This proves feasibility for this one source in the quadratic half-line cone. It makes no assertion that every valid source belongs to that cone.